RF E4 vs LightGBM E4: Class-Level Performance

Class	RF E4 Recall	RF E4 F1	LightGBM E4 Recall	LightGBM E4 F1	Better overall / interpretation
Normal	0.9468	0.9725	**0.9965**	**0.9982**	**LightGBM**
Short Circuit	**0.9996**	**0.9821**	0.4994	0.6561	**RF**
Resistive Degradation	0.9907	**0.9845**	**0.9930**	0.8980	**RF overall**
Open Circuit	1.0000	0.9990	0.9992	0.9990	**Essentially tied**
Building Shade	0.9911	0.8144	**0.9937**	**0.9594**	**LightGBM**

RF E4 is stronger for hardware fault-type identification, especially Short Circuit, while LightGBM E4 performs better on Normal and Building Shade and is substantially more resource-efficient.